

Exercise (Existence of perfect message authentication codes). *A deterministic message authentication code $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ is perfect if the success of impersonation and substitution attacks is bounded by $\frac{1}{|\mathcal{T}|}$ and $|\mathcal{K}| = |\mathcal{T}| \cdot |\mathcal{M}|$. It is easy to construct such codes for $|\mathcal{K}| = |\mathcal{M}| \cdot |\mathcal{M}|$ but it is harder for longer messages. Construct a perfect message authentication code $h : \mathbb{Z}_2^2 \times \mathbb{Z}_2^2 \rightarrow \mathbb{Z}_2$ or prove that no such codes can exist.*

Solution. Recall that a perfect message authentication code must be a universal hash function. Just start to fill the function table with corresponding restrictions.

Key	00	01	10	11
00				
01				
10				
11				